Jingcheng Liang

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Education

University of Toronto Mississauga Sept. 2024 – Apr. 2028 (expected)
Honours Bachelor of Science, Double Major in Computer Science and Statistics Mississauga, ON

- **CGPA: 4.0/4.0** | Dean's List Scholar (2025, 2026).
- **Selected A+ grades: Computer Science** – CSC108, CSC148, CSC209, CSC263, CSC311, CSC343; **Statistics** – STA256, STA258, STA260, STA302.

Teaching Experience

Teaching Assistant – Computer Science and Statistics Jan. 2026 – Present
University of Toronto Mississauga, Department of Mathematical and Computational Sciences Mississauga, ON

- Supported teaching in **CSC108, STA107, and STA258**, including lecture activities, statistics tutorials, office hours, and individual student assistance.
- Guided students through programming logic, Python debugging, probability concepts, and statistical problem-solving; adjusted explanations to their questions and misconceptions.
- Graded assessments with consistent rubrics and provided specific, constructive feedback to help students improve their reasoning and solutions.

Research and Software Development Experience

Research Developer – Generative AI Mastery Learning Platform Apr. 2026 – Present
University of Toronto Toronto, ON

- Develop LLM-supported mathematics learning tools and API-connected educational workflows with faculty and software developers.

Student Software Engineer – Office Hour Management System Apr. 2026 – Present
University of Toronto Mississauga, ON

- Build teaching-support software using **Next.js, TypeScript, PostgreSQL, and Prisma** for staff scheduling, check-ins, and office-hour sessions.
- Design data models, test application workflows, and help validate the system with developers and teaching staff in real course settings.

Research Assistant – Economics Research on Innovation and Firm Dynamics Apr. 2026 – Sept. 2026
University of Toronto Toronto, ON / Remote

- Built **Python** data-collection and LLM-assisted extraction pipelines; validated outputs against labelled samples and investigated errors to support reproducible research.

Selected Technical and Statistical Projects

MyShell – Unix-like Command Shell Winter 2026
Built an interactive Unix-like shell in **C/Linux** with pipelines, background jobs, process control, and signal handling; tested and debugged system calls and edge cases.

Federated Compute Coordinator – Scheduling and Simulation Fall 2026
Worked on coordinator lease-lifecycle design and test scenarios for simulated providers, job scheduling, timeouts, retries, and failures.

Work Performance Analysis Winter 2026
Used **R** to compare linear and quadratic regression models; evaluated residual assumptions, multicollinearity, and influential observations.

Artwork Classification Winter 2026
Developed a **Python/scikit-learn** classification pipeline with preprocessing, text features, and cross-validation on mixed data.

Skills

Programming / systems: Python, C, C++, Java, JavaScript, TypeScript, SQL, Git, Linux/Unix, PyTest, PostgreSQL

Statistics / data: R, R Markdown, Pandas, NumPy, SciPy, scikit-learn, regression, probability, inference, model evaluation

Teaching: Tutorials, lecture support, office hours, rubric-based grading, individualized feedback, programming help